

Problem 1

Given the following twist ξ , what is the homogeneous transformation matrix?

(a) $\xi = \begin{bmatrix} \mathbf{v}^\top & \boldsymbol{\Omega}^\top \end{bmatrix}^\top = \begin{bmatrix} 3/\pi & 0 & 0 & \pi/3 & 0 & 0 \end{bmatrix}^\top.$

(b) $\xi = \begin{bmatrix} \mathbf{v}^\top & \boldsymbol{\Omega}^\top \end{bmatrix}^\top = \begin{bmatrix} 0 & 0.5 & 0 & 0 & \pi/2 & 0 \end{bmatrix}^\top.$

(c) $\xi = \begin{bmatrix} \mathbf{v}^\top & \boldsymbol{\Omega}^\top \end{bmatrix}^\top = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \pi/3 \end{bmatrix}^\top.$

Part (a)

[Work not yet added: paste solution here]

Part (b)

[Work not yet added: paste solution here]

Part (c)

[Work not yet added: paste solution here]

Problem 2

Given the homogeneous transformation matrix,

$$\mathbf{T} = \begin{bmatrix} 0.5000 & -0.6124 & 0.6124 & 0.8415 \\ 0.6124 & 0.7500 & 0.2500 & 0.3251 \\ -0.6124 & 0.2500 & 0.7500 & -0.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- (a) What is the displacement?
- (b) What is the Quaternion that encodes the rotation?
- (c) What is the Angle-Axis that encodes the rotation?
- (d) What is the twist ξ that encodes the full transform?

Part (a): Displacement

[Work not yet added: paste solution here]

Part (b): Quaternion

[Work not yet added: paste solution here]

Part (c): Angle-Axis

[Work not yet added: paste solution here]

Part (d): Twist

[Work not yet added: paste solution here]

Problem 3

For the following give one pro and one con for using it as a representation,

- (a) Rotation and displacement separately.
- (b) Homogeneous transformation matrices.
- (c) Twist coordinates.

Part (a): Rotation and Displacement Separately

[Work not yet added: paste solution here]

Part (b): Homogeneous Transformation Matrices

[Work not yet added: paste solution here]

Part (c): Twist Coordinates

[Work not yet added: paste solution here]

Problem 4

Given the following two homogeneous transformations, use MATLAB to plot a linear interpolation between the two. Your plot should contain a minimum of 5 intermediate points and should have the orthogonal axis (columns of the rotation matrix) drawn at each point. *Hint:* Use the `quiver3` function.

$$\mathbf{T}_{\text{initial}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{T}_{\text{final}} = \begin{bmatrix} 0.5000 & -0.6124 & 0.6124 & 0.8415 \\ 0.6124 & 0.7500 & 0.2500 & 0.3251 \\ -0.6124 & 0.2500 & 0.7500 & -0.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Perform the interpolation using ZYZ angles and position.
- Perform the interpolation using Angle-Axis and position.
- Perform the interpolation using Quaternion and position (remember to enforce the length constraint on the quaternion before calculating the intermediate orientations).
- Perform the interpolation using twist.
- Comment on the similarities or differences.

Part (a): ZYZ Interpolation

[Work not yet added: paste solution here]

Part (b): Angle-Axis Interpolation

[Work not yet added: paste solution here]

Part (c): Quaternion Interpolation

[Work not yet added: paste solution here]

Part (d): Twist Interpolation

[Work not yet added: paste solution here]

Part (e): Comparison

[Work not yet added: paste solution here]

Appendix A: Code

Problem 1

Listing 1: Problem 1.py

Problem 2

Listing 2: Problem 2.py

Problem 3

Listing 3: Problem 3.py

Appendix B: Hand Calculations